LAB 9 : Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

Objectives:

- To configure a wireless router and establish a wireless network using Cisco Packet Tracer
- To connect wireless devices to the network and assign IP addresses using DHCP
- To verify secure wireless communication using WPA2 security settings

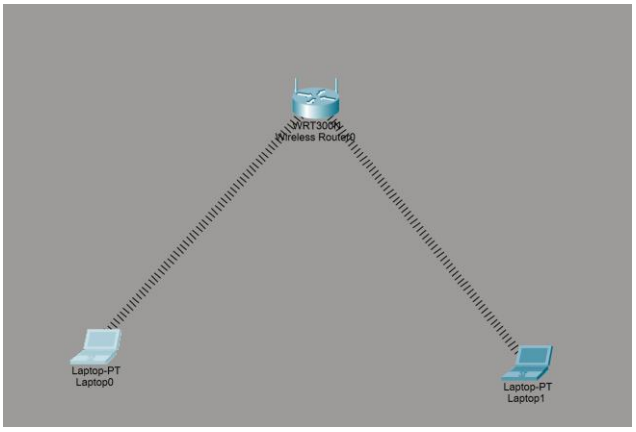
Theory:

A wireless network allows devices to exchange data over the air using radio signals instead of physical connections. In this experiment, a wireless router was set up to form a Wireless Local Area Network (WLAN) that offers network connectivity, automatic IP address distribution through DHCP, and secure access for wireless users.

The network is identified by a unique SSID, which enables wireless devices to locate and connect to it. To protect the network, WPA2-Personal security with AES encryption was implemented, ensuring that only authorized users with the correct passphrase can gain access. Clients join the network by selecting the SSID and providing the required security key.

Packet analysis was used to study communication between the wireless devices and the router. By using Simulation Mode in Cisco Packet Tracer, different packets such as DHCP, ARP, and data frames were observed. This analysis helped in understanding how IP addresses are assigned, how connections are established, and how data is transmitted wirelessly. Packet inspection also proved useful for verifying network behavior and identifying possible issues during operation.

Network Topology :



Procedure :

Wireless Router Configuration :

Step 1 : Basic Setup

1. Click on the Wireless Router.
2. Go to GUI → Setup.
3. Configure the following:
   - IP Address: 192.168.0.1
   - Subnet Mask: 255.255.255.0
   - DHCP Server: Enabled

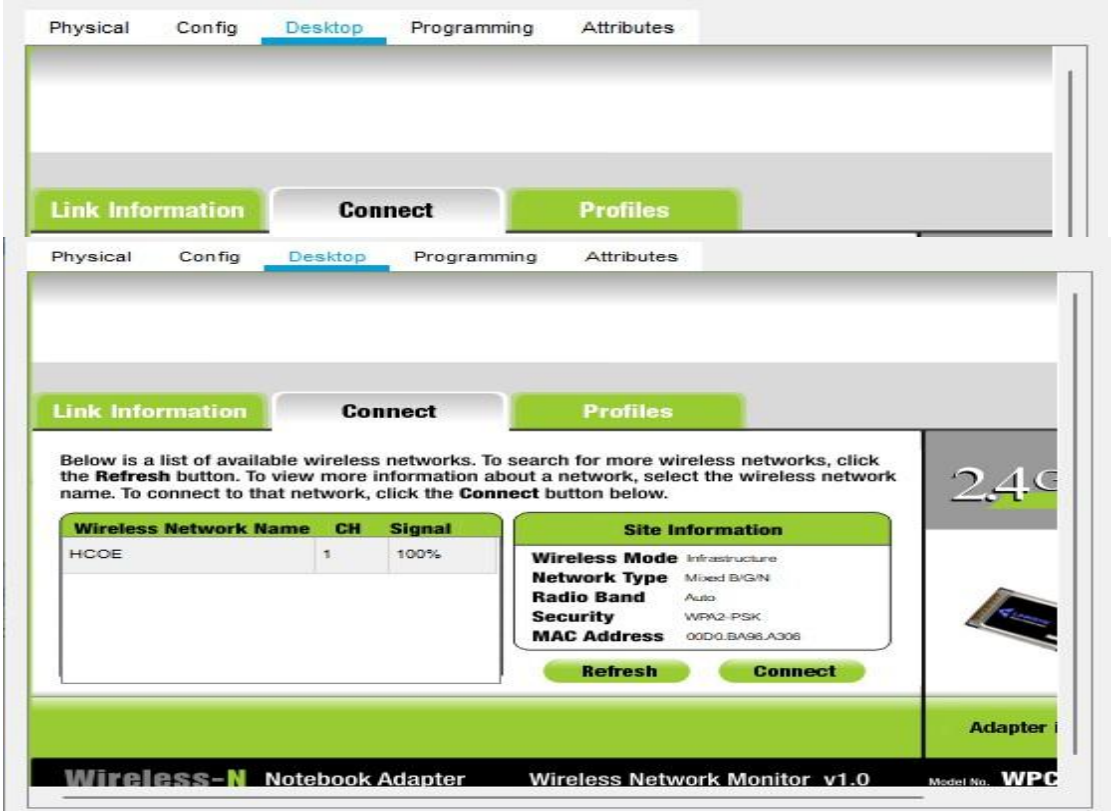
Step 2 : Wireless Configuration

1. Go to Wireless → Basic Wireless Settings.
2. Configure the following parameters:
3. Network Mode: Mixed
   - SSID: HCOE
   - Channel: Auto
4. Click Save Settings.

Step 3 : Wireless Security

1. Go to Wireless → Wireless Security.
2. Configure the following:
   - Security Mode: WPA2-Personal
   - Encryption: AES
   - Passphrase: hcoe@123
3. Save the configuration.

Observation :



Result:
A wireless network was set up successfully using the WRT300N router with proper security settings. The network was broadcast using the SSID HCOE and protected with WPA2-Personal encryption. The router's DHCP service, operating at the gateway address 192.168.0.1, automatically provided IP addresses to connected devices. Both Laptop0 and Laptop1 detected the wireless network with strong signal quality, authenticated using the correct passphrase, and communicated smoothly through the router.

Discussion & Conclusion:
This experiment focused on combining automatic IP configuration and wireless security within a single network. The use of DHCP reduced the need for manual network setup, making client configuration faster and more efficient. Implementing WPA2-Personal with AES encryption ensured that only authorized devices could access the network
This lab confirmed the importance of proper security and address management in wireless network deployment.